

Study Guide: System Analys

1. Overview of the Systems Development Life Cycle (SDLC)

The Systems Development Life Cycle (SDLC) is a structured, phased approach to developing high-quality information systems. It ensures that projects are managed effectively from the initial business need through to a fully operational solution.

The Five Phases of the SDLC

Phase	Primary Purpose
Planning	Identify the need for a new system and determine project viability(项目可行性) through a preliminary investigation(初步调查).
Analysis	Gather detailed requirements from end users to define exactly what the new system must do.
Design	Create the technical blueprints(技术蓝图), including hardware requirements, software architecture, and interface designs.
Development	Perform the actual programming, coding, and rigorous testing(严格测试) of the system components.
Implementation	Install the final system, migrate data, train users, and transition to live operations.

2. The System Request: Foundations of Planning

Every project begins with a **System Request**. This is a request for services from the Information Systems (IS) department to solve a business problem or improve a process. Requests can be **formal** (submitted via a standardized form or online ticket) or **informal** (a verbal instruction from management).

Reasons for System Requests

Reason	Description
Objectives Support	The current system cannot support new business goals, such as increasing sales or adding new modules like Customer Relationship Management (CRM).
System Problems / Lacking Functions	The existing system has errors, is too slow (poor performance), or misses features required to complete business processes.

Sources of System Projects

Source Group	Specific Examples / Drivers
Top Management	Strategic instructions to meet management needs or the requirement for new executive reports.
Users	Requests for new features to support job roles , such as handling "inward returns" in an inventory system(库存系统中的“入库退货”处理。).
System Problems	Inefficiencies(效率低下) or bugs reported by staff or observed by IS personnel(由信息系统人员观察).
External Sources	Pressure from new technology (RFID), government regulations (tax changes), or customer demands for online features.
IS Department	Recommendations based on poor system performance, discovered bugs, or a lack of documentation.

Types of Improvements

- **Performance:** Increasing speed, reducing transaction time, and supporting a higher volume of users.
 - **Functions:** Adding new capabilities (e.g., creating a Request for Quotation) or simplifying processes like overtime claims(加班索赔). *Example: Improving the share registration process for customers.*
 - **Quality of Reports:** Ensuring the information produced is **accurate, complete, relevant, useful, and timely**. *Example: Providing customers with up-to-date product brochures.*
 - **Effective Controls:** Improving weak security and access logs or streamlining "excessive" controls that slow down data input.
 - **Reduce Cost:** Lowering high manpower requirements for purchasing processes or reducing maintenance costs caused by buggy software.
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3. Preliminary Investigation and Feasibility Studies

The **Preliminary Investigation** occurs after a request is approved. It aims to determine if the proposed project is worth the investment. The primary output is the **Preliminary Investigation Report**, which management reviews to decide on further action.

The Three Types of Feasibility

Feasibility Type	Primary Concern	Key Issues to Address
Technical Feasibility	Hardware, software, and human requirements.	Can the system support the required number of users? Can it process large data inputs and maintain response times/deadlines?
Social and Operational Feasibility 运营可行性	Organizational fit and human impact (组织契合度和对人的影响。).	Does the project fit the organizational structure? Do employees have the skills? Will there be resistance to job changes? What are the retraining or redundancy costs?
Economic Feasibility	Financial viability (财务可行性).	Do the benefits outweigh the costs? Analysts use techniques like Return on Investment (ROI) , Internal Rate of Return (IRR) , and Net Present Value (NPV) to make the final "continue or halt" decision.

4. Economic Analysis: Costs and Benefits

Economic analysis ensures the project is a sound financial investment by quantifying the impact of the new system.

Project Costs (Development and Purchase)

Cost Group	Specific Elements
Hardware	Servers, client PCs, peripherals, and network infrastructure (cables, modems).
Software	Operating systems, DBMS, programming tools, and Application Software/Enterprise Systems (for purchase paths).
Development	System planning, analysis, design, software evaluation/selection , coding, and testing.
Implementation	Hardware/software installation, data migration, user training, and system changeover.

Intangible Costs

- **Learning Curve Costs:** The period where staff are slower and more prone to errors(速度较慢且更容易出错) while mastering the new system.
- **Dysfunctional Costs:** Negative staff behaviors, such as dissatisfaction or decreased morale(士气低落), resulting from the change.
- **Incompatibility Costs:** The expense of replacing old systems or building data conversion facilities to link new and old systems.
- **Opportunity Costs:** The loss of potential gains from other projects because funds were directed to this system.

Tangible vs. Intangible Benefits

Tangible (Quantifiable) Benefits 切实(可量化)的收益	Intangible (Non-Quantifiable) Benefits 无形(无法量化)收益
Reduced staff costs (fewer personnel).	Improved company image and customer satisfaction.
Reduced operating costs (less office/warehouse space).	Improved staff morale, motivation, and retention.
Reduced stock holding costs (less obsolescence).	Better information accessibility and faster decision-making.
Increased sales and productivity.	Greater system reliability and security.

5. System Analysis: Fact-Gathering Techniques

System Analysis, or **Requirements Analysis**, is the process of defining exactly what functionalities the new system must provide. This information is documented in the **System Requirements Specification**.

Comparison of Fact-Gathering Techniques

Method	Description / Meaning	Advantages (Pros)	Disadvantages (Cons)
Interviews	Formal face-to-face meetings to ask users about requirements.	Builds rapport(建立融洽关系); allows observation of body language; clarifies confusing facts(澄清令人困惑的事实).	Costly and time-consuming; requires high interpersonal skills(需要具备较强的人际交往能力).
Questionnaires	Standardized forms (closed and open questions) sent to users.	Efficient for large or geographically distributed groups; allows anonymity(允许匿名).	Difficult to design without bias(难以设计出无偏见的方案); cannot clarify answers; low return rates.
Observation	Watching users perform procedures and interact with data in real-time.	Provides a reality check; reveals environmental issues (noise, space).	Time-consuming; Hawthorne effect (users act differently when watched).
Document Analysis	Reviewing existing reports, manuals, and organization charts.	Clear understanding of formal processes and existing data structures.	May not reflect informal "shortcuts" used by staff in daily work.
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JAD	Interactive workshops(互动研讨会) where IT and users work together as a group.	Efficient exploration of requirements; immediate group discussion.	Requires a highly skilled project leader to manage group dynamics.
Prototyping	Creating a small-scale working model for user testing.	High productivity; allows immediate feedback on the user interface.	Can lead to unrealistic expectations regarding full system delivery.

Research and Prototyping

- **Research:** Essential when no existing system or documentation is available. It involves reviewing journals, attending professional seminars, and visiting similar sites.
 - **Prototyping Productivity:** By using **4GL** or **CASE tools**, analysts can quickly develop models that can be evolved into complete application programs in a much shorter time frame.
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6. Functional and Non-Functional Requirements

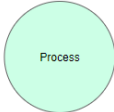
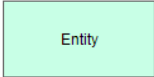
Requirements define the specifications that the new system must satisfy.

Functional Requirements	Non-Functional Requirements
Specific activities, tasks, or services the system must perform to support business processes.	Constraints, standards, or quality attributes the system must satisfy (security, legal, performance).
Examples: Create Purchase Order, Receive RFQ, Answer Enquiries, Calculate Gross Pay.	Examples: Audit Trails (transaction logs), Data Protection/Legal compliance, Backups.

7. Fact Recording: Data Flow Diagrams (DFDs)

Data Flow Diagrams are graphical models used to document the logical flow of data through a system.

DFD Components and Symbols (TAR UMT Recommended)

Component Name	Symbol Description	Simple Definition
Data Flow	Arrow (straight or curved) 	The path by which specific pieces of data move between parts of the system.
Process	Perfect Circle 	The transformation of incoming data flows into outgoing data flows.
Data Store	Open-ended Rectangle 	A file or table where information is at rest. Must have a Reference Number .
Entity	Rectangle 	A person, department, or external system that sends or receives information.

DFD Levels and Abstraction

- **Context Diagram (Level 0):** Represents the entire system as a single process (numbered 0). It shows the system boundary and external entities but **never** shows data stores.
- **Diagram 0:** An overview showing the major high-level processes, data stores, and entities.
- **Diagram 1:** A detailed "Explosion" of a single process from Diagram 0.
- **Explosion:** The process of breaking a complex process into deeper levels of detail until reaching **functional primitives** (the lowest level of detail where a process cannot be broken down further).

The Rules of DFDs

1. **Transformation:** Processes must have at least one input and one output; the output must differ from the input.
2. **Naming:** Processes must be labeled with an active verb and a noun (e.g., "Prepare Invoice").
3. **Data Flow Direction:** Arrows must always start or end at a process. They cannot connect an entity directly to a data store or another entity.
4. **Labeling:** Every data flow arrow must be labeled. There is no requirement to label arrows that go in and out of data stores.
5. **Data Stores:** Used only for data at rest and must include a reference number.
6. **Balancing:** The number of inflows and outflows must remain consistent when moving between levels (e.g., from a Context Diagram to Diagram 0).

Common DFD Errors

Error Name	Definition	Example
Miracle	A process that has output data flows but no input data flows.	Apply insurance premium: The process produces a policy number/payment amount with no inputs.
Black Hole	A process that has input data flows but produces no output.	Calculate gross pay: Hours worked and pay rate enter the process, but nothing leaves.
Gray Hole	A process where the inputs are insufficient to produce the output.	Calculate grade: The process only has "Date of Birth" as an input but tries to output a "Final Grade."